Assignment-17.2

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TaskDescription–1(StudentAcademicDataCleaning)

- **Task:**

UseAIassistancetogenerateaPythonscriptthatcleansand prepares a student academic dataset for analysis.

Instructions:

- **Handlemissingvaluesinmarks,attendance,anddepartment.**
- **Removeduplicatestudentrecordsbasedonstudent ID.**
- **Standardizetextfieldssuchasdepartmentnames.**
- **Encodecategoricalvariableslikegenderanddepartment.**

Sample Code:

```
importpandasas pd
data =pd.read_csv("students.csv")
print(data.head())
```

ExpectedOutput:

- **AcleanedandencodedPandasDataFramesuitableforacademic performanceanalysis.**

Prompt:

Generate Python code using pandas to clean a student dataset. Handle missing values, remove duplicates based on student ID, standardizedepartmentnames,andencodecategoricalvariables**. Code:**

task1.py ● students.csv BLACKBOX

task1.py > ...

```python
#Generate Python code using pandas to clean a student dataset.
#Handle missing values, remove duplicates based on student ID,
#standardize department names, and encode categorical variables.
import pandas as pd
# Load data
data = pd.read_csv("students.csv")
# Handle missing values (FIXED)
data['marks'] = data['marks'].fillna(data['marks'].mean())
data['attendance'] = data['attendance'].fillna(data['attendance'].median
data['department'] = data['department'].fillna("Unknown")
# Remove duplicates
data = data.drop_duplicates(subset='student_id')
# Standardize text
data['department'] = data['department'].str.lower().str.strip()
# Encode categorical variables
data['gender'] = data['gender'].map({'Male': 0, 'Female': 1})
data = pd.get_dummies(data, columns=['department'])
print(data)
```

PROBLEMS OUTPUT TERMINAL ··· powershell

```
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17> python task1.py
   student_id     name  ...  department_ece  department_unknown
0           1    Alice  ...           False               False
1           2      Bob  ...            True               False
2           3  Charlie  ...           False               False
3           4    David  ...           False                True

[4 rows x 8 columns]
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17>
```

students.csv > data

```csv
student_id,name,marks,attendance,department,gender
1,Alice,85,90,CSE,Female
2,Bob,,80,ECE,Male
3,Charlie,78,,CSE,Male
4,David,92,95,,Male
1,Alice,85,90,CSE,Female
```

Explanation:

Thedatasetwasloadedusingpandas.Missingvaluesinmarks,attendance,and department columns were handled using mean, median, and default values respectively.Duplicatestudentrecordswereremovedbasedonstudent ID.The department names were standardized by converting them into lowercase. Categoricalvariablessuchasgenderanddepartmentwereencodedinto numerical format using mapping and one-hot encoding techniques.

TaskDescription–2(FinancialTransactionDataPreprocessing)

- **Task:**

PreprocessfinancialtransactiondatausingAI-generatedPython code.

Instructions:

- **Converttransactiondatecolumnstodatetimeformat.**
- **Createderivedfeaturessuchastransactionmonthandyear.**
- **Normalizethetransactionamountcolumn.**
- **Identifyandhandleextremetransactionvalues.**

Sample Code:

```
importpandasas pd
fromsklearn.preprocessingimportMinMaxScaler
transactions = pd.read_csv("transactions.csv")
print(transactions.info())
```

ExpectedOutput:

- **Atransformeddatasetwithnormalizedvaluesandnewly generated time-based features.**

Prompt:

GeneratePythoncodetopreprocessfinancialtransactiondata.
Convert date columns, extract month/year, normalize amount,
and detect extreme values.
Code:

py | students.csv | transactions.csv | task2.py | BLACKBOX

task2.py > ...

```python
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
# Load data
transactions = pd.read_csv("transactions.csv")
# Convert date column
transactions['date'] = pd.to_datetime(transactions['date'])
# Create new features
transactions['month'] = transactions['date'].dt.month
transactions['year'] = transactions['date'].dt.year
# Normalize amount
scaler = MinMaxScaler()
transactions['amount_normalized'] = scaler.fit_transform(transact
# Remove outliers using IQR
Q1 = transactions['amount'].quantile(0.25)
Q3 = transactions['amount'].quantile(0.75)
IQR = Q3 - Q1
transactions = transactions[
    (transactions['amount'] >= Q1 - 1.5 * IQR) &
    (transactions['amount'] <= Q3 + 1.5 * IQR)
]
# Display output
```

PROBLEMS OUTPUT TERMINAL ··· powershell

```
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17> python task2.py
   transaction_id       date  amount  month  year  amount_normalized
0               1 2023-01-15    5000      1  2023           0.047141
1               2 2023-02-20   15000      2  2023           0.147442
2               3 2023-03-10     300      3  2023           0.000000
4               5 2023-05-18     700      5  2023           0.004012
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17>
```

task1.py | students.csv | transaction

transactions.csv > data

```csv
transaction_id,date,amount
1,2023-01-15,5000
2,2023-02-20,15000
3,2023-03-10,300
4,2023-04-05,100000
5,2023-05-18,700
```

Explanation:
The transaction dataset was loaded and the date column was converted into datetimeformat.Newfeaturessuchasmonthandyearwereextractedfromthe date column. The transaction amount was normalized using MinMaxScaler to bringallvaluesintoastandardrange.Outlierswerredetectedandremovedusing the Interquartile Range (IQR) method to improve data quality.

Task3–(EnvironmentalSensorDataPreparation)
• Task:
CleanandpreprocessenvironmentalsensordatausingAI-assisted scripts.
Instructions:
• Handlemissingvaluesintemperature,humidity,and air_quality_index.
• Scalenumericalfeaturesusingstandardscaling.
• Encodecategoricalfieldssuchassensor_status.
• Splitthedatasetintotrainingandtestingsubsets.
Sample Code:
importpandasaspd

fromsklearn.model_selectionimporttrain_test_split
data = pd.read_csv("sensor_data.csv")
ExpectedOutput:

- **A fully preprocessed dataset ready for predictive modeling tasks.**

_Prompt:
GeneratePythoncodetocleansensor data.
Handlemissingvalues,scalenumericalfeatures,
encode categorical fields, and split dataset**.**
Code:

task3.py

```python
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
# Load data
data = pd.read_csv("sensor_data.csv")
# Handle missing values
data['temperature'] = data['temperature'].fillna(data['temperatur
data['humidity'] = data['humidity'].fillna(data['humidity'].mean(
data['air_quality_index'] = data['air_quality_index'].fillna(data
# Encode categorical variable
data['sensor_status'] = data['sensor_status'].map({'active': 1, 'inact
# Scale numerical features
scaler = StandardScaler()
num_cols = ['temperature', 'humidity', 'air_quality_index']
data[num_cols] = scaler.fit_transform(data[num_cols])
# Split dataset
train_data, test_data = train_test_split(data, test_size=0.2, ran
```

PROBLEMS OUTPUT TERMINAL powershell

```
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17> python task3.py
TRAIN DATA:
    temperature  humidity  air_quality_index  sensor_status
4      1.621266       1.5                1.5              0
2      0.000000      -1.5               -1.5              0
0     -0.540422      -0.5               -0.5              1
3     -1.405098       0.5                0.0              1

TEST DATA:
    temperature  humidity  air_quality_index  sensor_status
1      0.324253       0.0                0.5              1
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17>
```

Explanation:
Missingvaluesintemperature,humidity,andairqualityindexwerehandledusing mean imputation. The categorical variable sensor_status was encoded into numerical values. Numerical features were standardized using StandardScaler to ensure a uniform scale. The dataset was then split into training and testing sets using train_test_split for further machine learning tasks.

Task4–(Real-TimeApplication:SmartCityTrafficDataCleaning)

- **Scenario:**

Asmartcitysystemcollectstrafficdatafrommultiplesensors, resulting in noisy and inconsistent records.

Requirements:

- **Standardizeroadandlocationnames.**
- **Fillmissingtrafficdensityvaluesusingappropriatestatistical methods.**
- **Removeduplicatesensorreadings.**
- **Normalizespeedandvehiclecountmetrics.**
- **Generateabriefsummarycomparingdatasetqualitybeforeand aftercleaning**

SampleCode:

```
importpandasas pd
traffic = pd.read_csv("traffic_data.csv")
print(traffic.describe())
```

ExpectedOutput:

- **Acleanedandnormalizedtrafficdatasetwithanaccompanying data-quality improvement summary.**

Prompt:
Generate Python code to clean sensor data.
Handlemissingvalues,scalenumericalfeatures,
encode categorical fields, and split dataset.
Code:

task4.py > ...

```python
#Standardize road and location names, handle missing traffic dens
#remove duplicate records, normalize speed and vehicle count,
#and compare dataset quality before and after cleaning.
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
# Load data
traffic = pd.read_csv("traffic_data.csv")
# Before cleaning summary
print("===== BEFORE CLEANING =====")
print(traffic.describe(include='all'))
# Standardize text fields
traffic['road'] = traffic['road'].str.lower().str.strip()
traffic['location'] = traffic['location'].str.lower().str.strip()
# Handle missing values
traffic['traffic_density'] = traffic['traffic_density'].fillna(tr
# Remove duplicates
traffic = traffic.drop_duplicates()
# Normalize numerical features
scaler = MinMaxScaler()
traffic[['speed', 'vehicle_count']] = scaler.fit_transform(
    traffic[['speed', 'vehicle_count']]
)
# After cleaning summary
print("\n===== AFTER CLEANING =====")
print(traffic.describe(include='all'))
```

Output:

```
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17> python task4.py
===== BEFORE CLEANING =====
           road   location  traffic_density       speed  vehicle_count
count         6          6         4.000000    6.000000       6.000000
unique        5          5              NaN         NaN            NaN
top     Highway  Outskirts              NaN         NaN            NaN
freq          2          2              NaN         NaN            NaN
mean        NaN        NaN        60.000000   66.666667     248.333333
std         NaN        NaN        24.494897   26.770631     118.729384
min         NaN        NaN        30.000000   40.000000     150.000000
25%         NaN        NaN        45.000000   47.500000     165.000000
50%         NaN        NaN        65.000000   57.500000     190.000000
75%         NaN        NaN        80.000000   90.000000     350.000000
max         NaN        NaN        80.000000  100.000000     400.000000

===== AFTER CLEANING =====
             road     location  traffic_density     speed  vehicle_count
count           5            5         5.000000  5.000000        5.00000
unique          3            3              NaN       NaN            NaN
top     main road  city center              NaN       NaN            NaN
freq            2            2              NaN       NaN            NaN
mean          NaN          NaN        56.000000  0.333333        0.27200
std           NaN          NaN        18.165902  0.395285        0.41415
min           NaN          NaN        30.000000  0.000000        0.00000
25%           NaN          NaN        50.000000  0.083333        0.04000
unique          3            3              NaN       NaN            NaN
top     main road  city center              NaN       NaN            NaN
freq            2            2              NaN       NaN            NaN
mean          NaN          NaN        56.000000  0.333333        0.27200
```

Explanation:
Missingvaluesintemperature,humidity,andairqualityindexwerehandledusing mean imputation. The categorical variable sensor_status was encoded into numerical values. Numerical features were standardized using StandardScaler to ensure a uniform scale. The dataset was then split into training and testing sets using train_test_split for further machine learning tasks.

Task5–(SocialMediaTextDataCleaningandFeatureExtraction)

- **Task:**

UseAI-assistedPythonscriptingtocleanandpreprocesssocial media text data for sentiment analysis.

Instructions:

- **Removeduplicatepostsandnulltextentries.**
- **CleantextbyremovingURLs,specialcharacters,andextra spaces.**
- **Converttexttolowercaseandperformbasictokenization.**
- **Removestopwordsandperformstemmingorlemmatization.**
- **Generatebasictextfeaturessuchaswordcountandsentiment score.**

SampleCode:

importpandasaspd

import re

posts = pd.read_csv("social_media_posts.csv")

print(posts.head())

ExpectedOutput:

Acleanedtextdatasetwithextractedfeaturessuitableforsentiment analysis models.

Prompt:

GeneratePythoncodetocleansocialmediatextdata.
Remove duplicates and null values, clean text, tokenize,removestopwords,andextractfeatureslikesentiment.

Output:

task5.py > ...

```python
import pandas as pd
import re
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.sentiment import SentimentIntensityAnalyzer
# Download required resources (first time only)
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('vader_lexicon')
# Load data
posts = pd.read_csv("social_media_posts.csv")
# Remove duplicates and null values
posts = posts.drop_duplicates()
posts = posts.dropna(subset=['text'])
# Clean text function
def clean_text(text):
    text = re.sub(r"http\S+", "", text)          # remove URLs
    text = re.sub(r"[^a-zA-Z ]", "", text)       # remove special
    text = text.lower().strip()                 # lowercase
    return text
posts['clean_text'] = posts['text'].apply(clean_text)
# Tokenization and stopword removal
```

```
[nltk_data] Downloading package vader_lexicon to
[nltk_data]     C:\Users\dell\AppData\Roaming\nltk_data...
[nltk_data]   Package vader_lexicon is already up-to-date!
                        text  ... sentiment
0  I love AI! https://ai.com  ...    0.6369
1             This is bad!!!  ...   -0.5423
2      Amazing experience :)  ...    0.5859
5        AI is the future!!!  ...    0.0000
6      Worst service ever!!!  ...   -0.6249

[5 rows x 5 columns]
PS C:\Users\dell\OneDrive\Desktop\AI\Lab17>
```

Explanation:
The dataset was cleaned by removing duplicate and null entries. Text preprocessing included removing URLs, special characters, and converting text tolowercase.Tokenizationwasperformedusingsimplesplitting,andstopwords were removed to retain meaningful words. Additional features such as word count and sentimentscore were extracted using sentiment analysis techniques.